LECTURE 5

NEURAL NETWORKS AS KNOWLEDGE BASES: INTERPOLATION, GROKKING, AND LOCATION ENCODERS

Geospatial Representation Learning

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Learning Outcomes

Lecture

- Explain how modern overparameterized models challenge the classical bias-variance view of machine learning.
- Describe the interpolation regime, double descent, and grokking at a conceptual level.
- Interpret neural networks as implicit representations or neural knowledge bases that can store information in their parameters.
- Explain how implicit neural representations can encode spatial, temporal, or spatiotemporal data.
- Describe location encoders as neural models that map geographic coordinates and context to learned representations.
- Compare geospatial interpolation with learned location-based representation models.
- Explain the basic idea of SatCLIP-/GeoCLIP-like self-supervised geolocalization objectives.

Lab

- Query or construct a small geospatial embedding database.
- Use location-based embeddings for similarity search, prediction, or spatial interpolation.
- Implement a simple location encoder or use an existing pretrained location representation.
- Compare location-based representations against conventional coordinate or hand-crafted features.

PRACTICAL 5

LOCATION ENCODERS AND EMBEDDING DATABASES

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